Practice Problems #25

1. In the following, V is a nonzero finite-dimensional vector space, and the vectors listed belong to V.
   True or False. Justify:

   (a) If there exists a set $\{v_1, \cdots, v_p\}$ that spans V, then dim $V \leq p$.

   (b) If there exists a linearly independent set $\{v_1, \cdots, v_p\}$ in V, then dim $V \geq p$.

   (c) If dim$V = p$, then there exists a spanning set of $p + 1$ vectors in V.

   (d) If there exists a linearly dependent set $\{v_1, \cdots, v_p\}$ in V, then dim $V \leq p$.

2. Prove that the first four Hermite polynomials $1, 2t, -2+4t^2$, and $-12t+8t^3$ are linearly independent in $\mathbf{P_3}$ using Coordinate mapping idea. Then, explain why the set $S = \{1, 2t, -2 + 4t^2, -12t + 8t^3\}$ must be a basis of $\mathbf{P_3}$.

Answers or Hints are on page 2.

#1(a) T, (b) T, (c) T (justify them!), (d) F (create a counterexample)

#2: Mimic what we did in class. What is $\dim \mathbf{P_3}$? How many vectors in S?